BizView

MSME Business Intelligence & Loan System

Team: Finnovator

1. Introduction

1.1 Background of Case Study

Micro, Small & Medium Enterprises (MSMEs) form the backbone of many local economies but often lack affordable, data-driven tools to manage sales, inventory, forecasting, and access to credit. BizView is a hackathon prototype built to help MSME owners and small lenders make faster, more accurate decisions using automation and lightweight AI models. It combines a modern web frontend (Next.js) with a Python ML backend (FastAPI) to deliver dashboard analytics, demand forecasting, anomaly detection, and a loan pre-eligibility assessment workflow.

1.2 Chosen SDG

Primary: SDG 8 — *Decent Work and Economic Growth* (supporting MSME resilience, formalisation, and sustainable economic opportunities).
Secondary: SDG 9 — *Industry, Innovation and Infrastructure* (by improving digital infrastructure for small businesses).

1.3 Problem Statement

MSMEs in Malaysia face significant barriers in traditional business management and access to finance. Key challenges include:

- Limited access to **real-time business analytics** and **AI-driven insights** for tracking revenue, profit margins, product performance, demand forecasting, anomaly detection, and promotion strategies.

- Difficulty obtaining loans due to opaque eligibility criteria and lack of standardized business scoring.

- Inefficient processes for inventory management, sales tracking, point-of-sale operations, and strategic planning (e.g., testing discount/marketing scenarios).

These issues hinder data-driven decision-making, cost optimization, growth planning, and overall financial inclusion for MSME owners. Financial institutions also lack tools for effective risk assessment, market intelligence, and automated borrower-lender matching.

1.4 Objectives

1. Provide real-time KPIs for revenue, profit, transactions, and inventory.

2. Offer accurate demand forecasting including considering holiday, seasons, product type and scenario simulations so owners can plan stock and promotions.

3. Detect unusual transactions that may indicate errors or fraud.

4. Improve loan access via an intelligent scoring system (Random Forest Classifier) that pre-qualifies businesses and provides personalized loan recommendations.

5. Deliver a modern, usable web UI to improve MSME adoption.

2. Review of Existing Solutions

2.1 Common Categories of Existing Solutions:

- Cloud accounting & bookkeeping platforms (in market examples such as QuickBooks and Xero).
- POS and payments providers (example: Square).
- Lending marketplaces and credit scoring providers.

2.2 Gaps / Limitations Addressed by BizView

- Existing accounting tools focus on bookkeeping but often lack simple, integrated ML-based loan pre-eligibility tailored to MSMEs.
- Many POS systems do not provide demand forecasting or anomaly detection in an accessible format for non-technical owners.
- Lenders often request manual proofs and paperwork; BizView produces standardised, data-driven loan signals owners can present quickly.

3. Methodology

3.1 Technical Track — Architecture & Tech Stack

3.1.1 Frontend

- **Next.js 14** with App Router - React framework
- **TypeScript** - Type safety and better DX
- **TailwindCSS** + **shadcn/ui** - Modern, accessible UI components
- **Recharts** - Data visualization
- **Axios** - HTTP client for API communication

3.1.2 Backend

- **FastAPI** - High-performance async web framework
- **Pandas** & **NumPy** - Data manipulation and analysis
- **Scikit-learn** - Random Forest (loan scoring), Isolation Forest (anomaly detection)
- **Statsmodels** - ARIMA forecasting (statistical time-series)
- **Session-based prototype persistence** (session state / CSV import)

3.1.3 Machine Learning Models

- **Random Forest Classifier** - Loan eligibility scoring
- **Isolation Forest** - Anomaly detection in sales patterns
- **ARIMA** - Time series forecasting for demand prediction

3.2 Implementation Process

1. **Requirement analysis & UX**: Defined pages (Dashboard, POS, CSV Upload, Analytics, Forecast, Simulation, Loan Center).

2. **Data model design**: User profile, products table, sales table (product, quantity, date, revenue).

3. **Backend endpoints**: Create REST APIs for business profile, product CRUD, sales ingestion, CSV import, analytics, forecast, loan-scoring.

4. **ML & analytics pipeline**:

    - Preprocessing: aggregate sales to daily revenue, compute revenue stability (standard deviation & mean), profit margin, debt ratio.

    - Loan scoring:

        - ***Rule-based mode* (initial)**: weighted components (Revenue Stability 0.4, Profit Margin 0.3, Cash Flow Ratio 0.2, Debt Ratio inverse 0.1).

        - ***Auto ML mode***: train RandomForestClassifier once ≥ 20 sales records exist.

    - Forecasting: ARIMA(1,1,1) for 7+ day series
    - Anomaly detection: IsolationForest with 10% contamination.

5. **Frontend & UX**: Implement dashboard with KPI cards, charts for monthly revenue, top products, inventory status, and interactive simulation pages for discounts and marketing boost scenarios.

6. **Testing & debugging**: Local testing of endpoints, manual CSV uploads, UX checks.

4. Results

4.1 Prototype Deliverables:

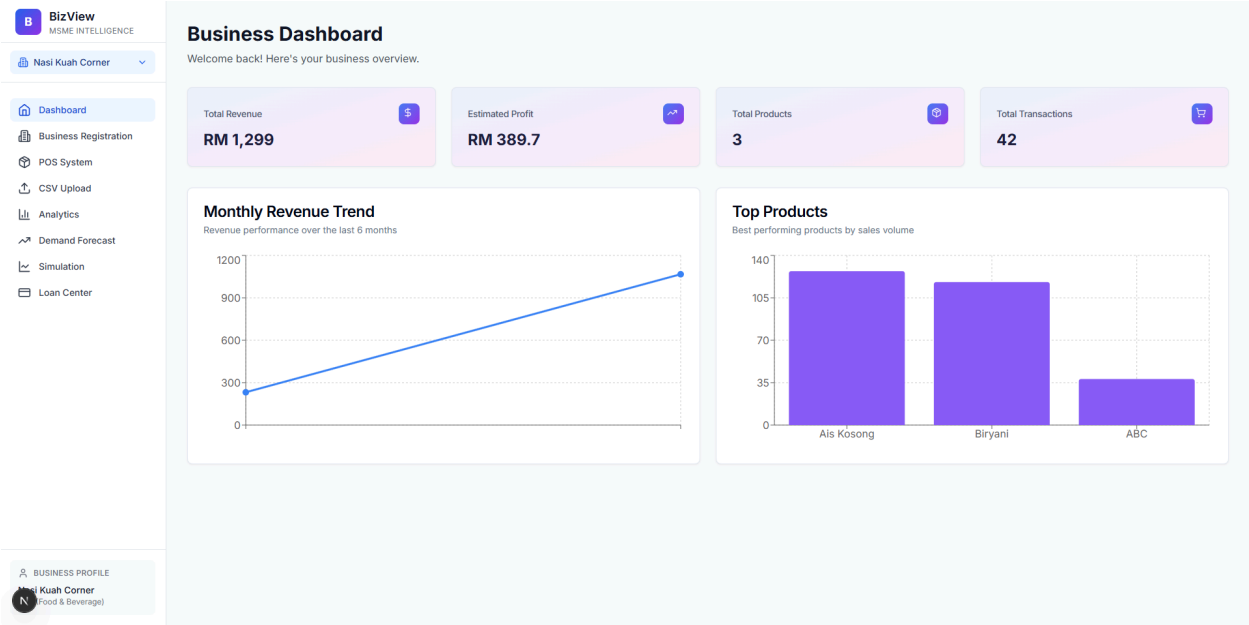
- FastAPI backend with the following endpoints and ML logic implemented:
    - **POST /api/business** — Create and update business profile
    - **POST /api/products** — Add product
    - **POST /api/sales** — Record sale
    - **POST /api/upload-csv** — Ingest historical sales
    - **GET /api/analytics** — Compute KPIs and top products
    - **GET /api/forecast** — Return 30-day ARIMA forecasts + confidence intervals
    - **GET /api/anomalies** — List flagged sales
    - **GET /api/loan-score** — Compute loan score and eligibility tier

Did we meet objectives?

- **Core objectives:** (real-time KPIs, demand forecasting, anomaly detection, loan pre-eligibility logic and POS flow) were met at prototype level . The prototype is ready for demonstration and further polish.

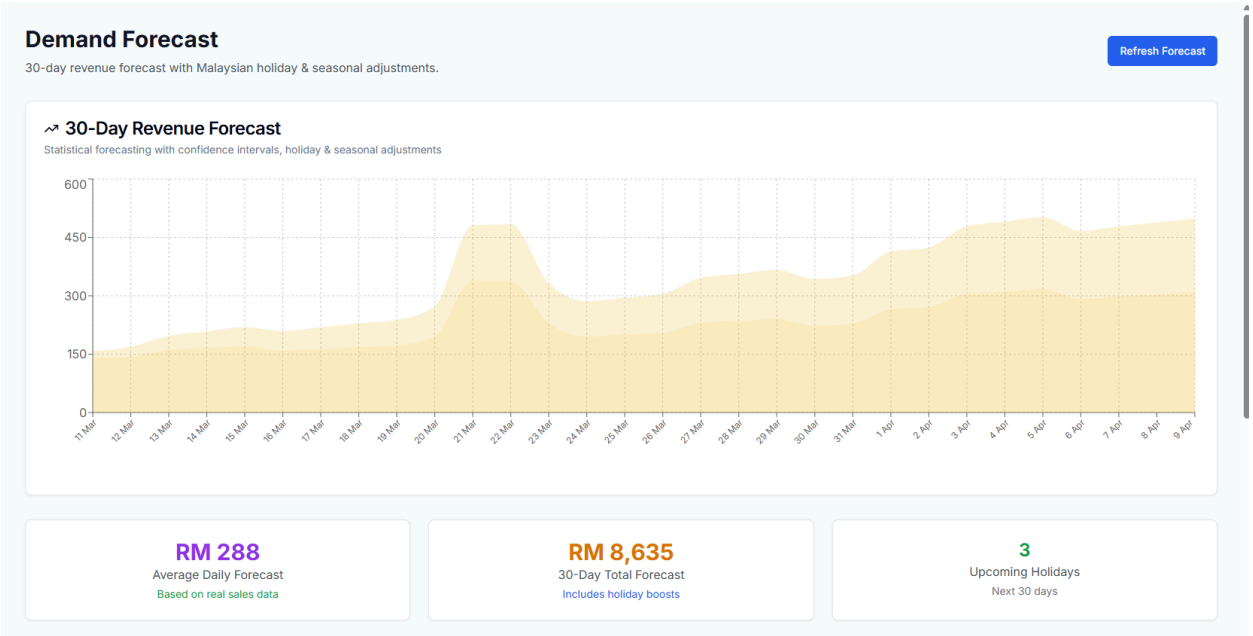
5. Prototype Showcase

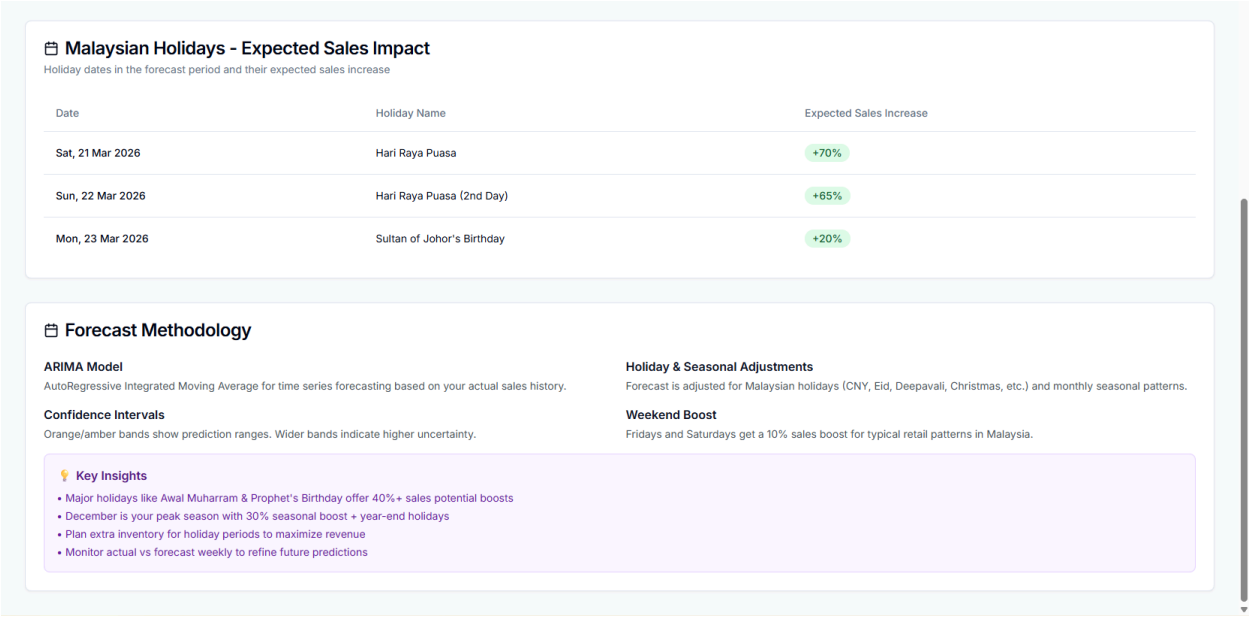
1. **Business Intelligence Dashboard**
    - Real-time KPIs:
        - Revenue
        - Profit margins
        - Transactions
        - Top-selling products



2. Demand Forecasting

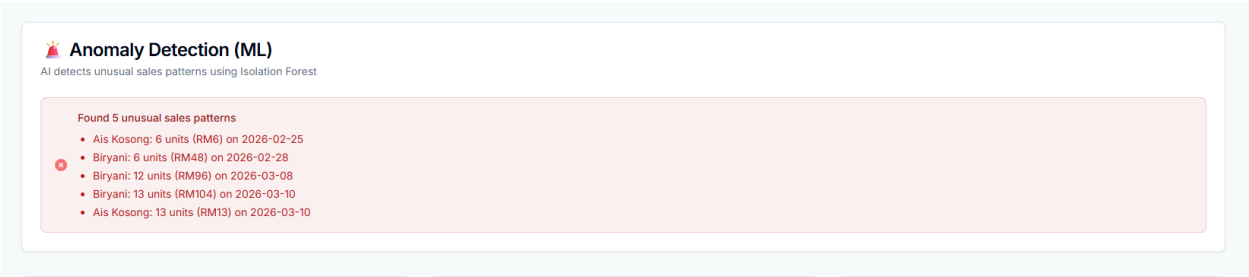
- AI predicts future sales trends using ARIMA time-series forecasting.





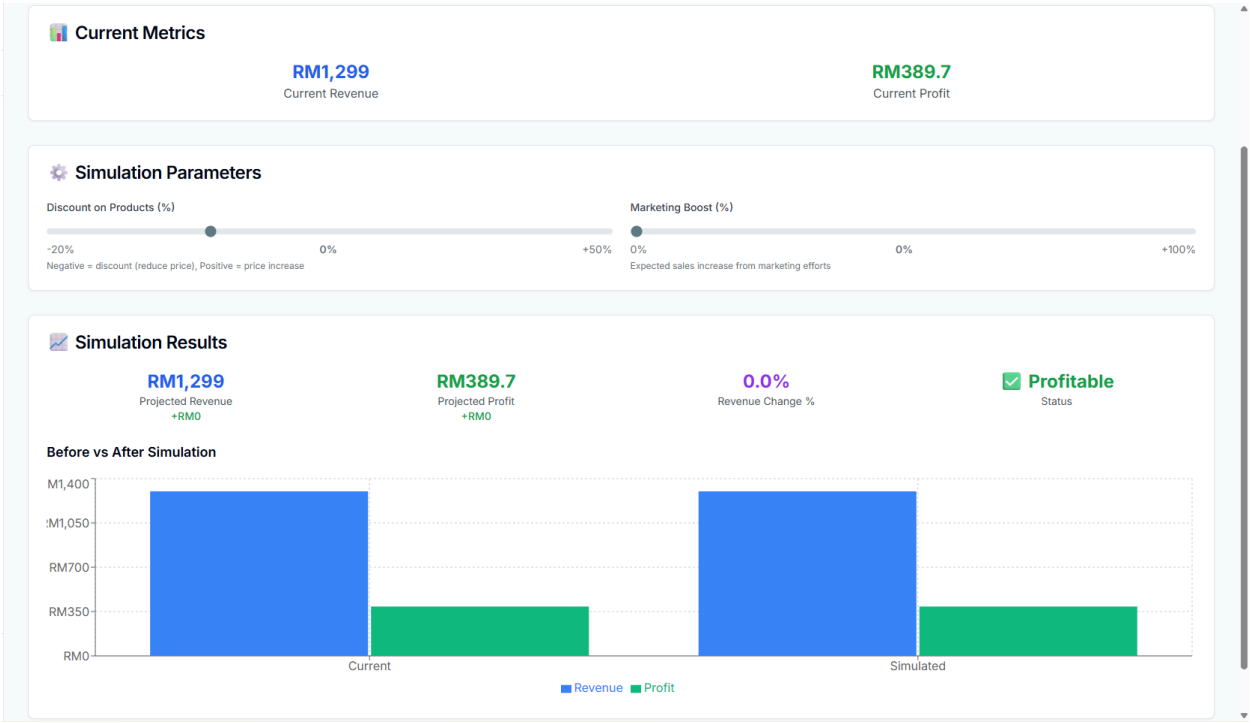
3. Anomaly Detection

- Isolation Forest identifies suspicious or unusual sales patterns.



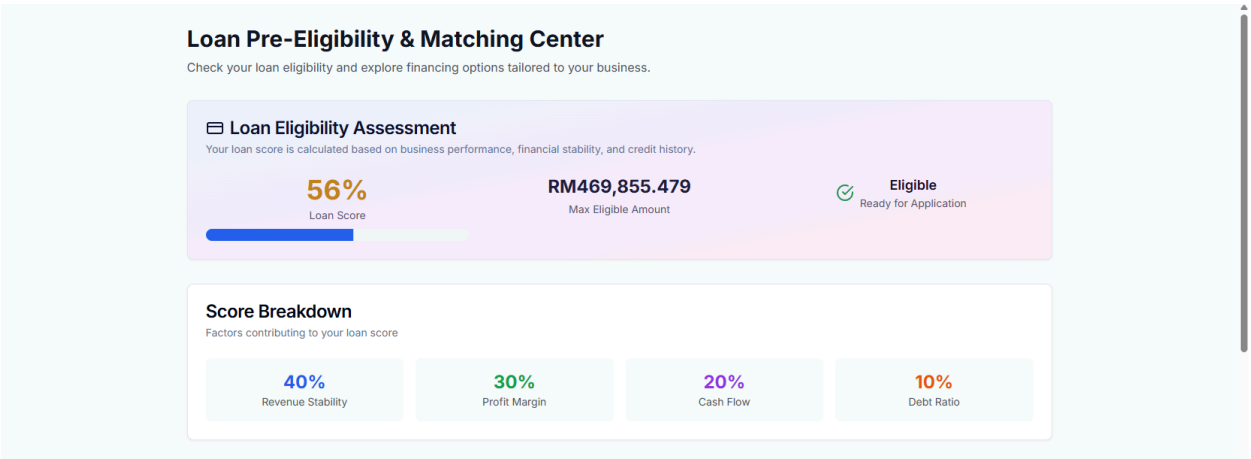
4. Scenario Simulation

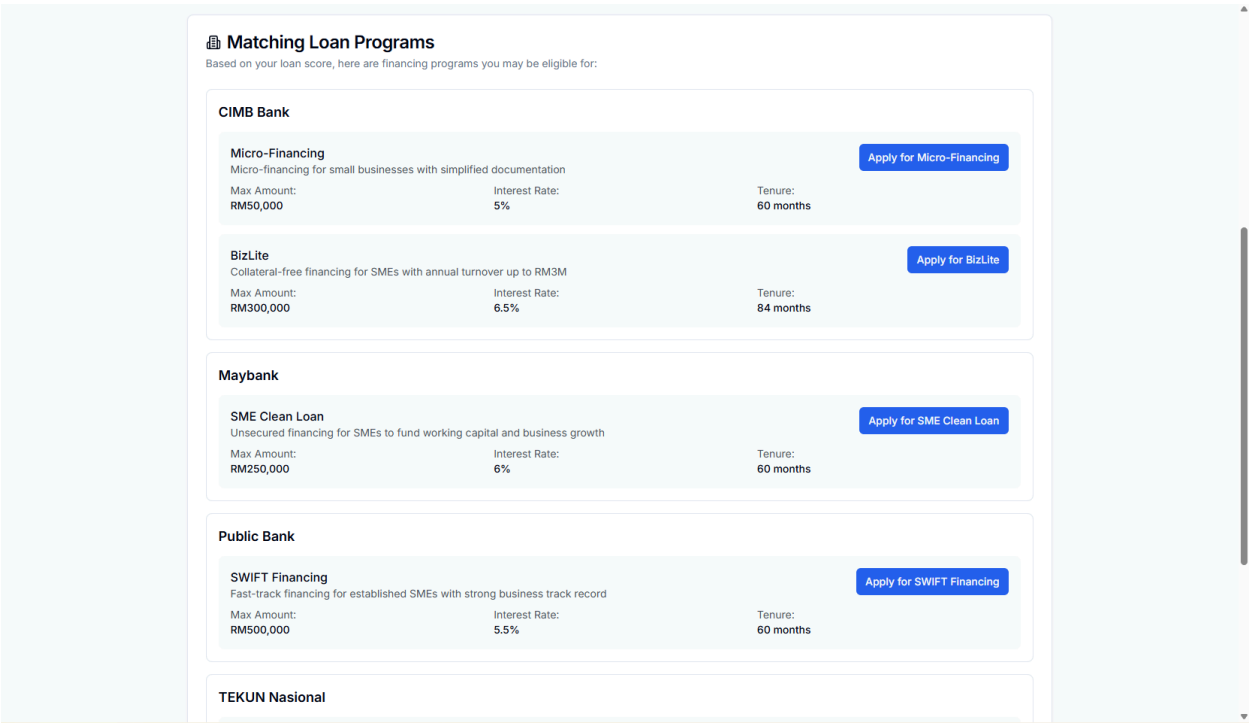
- Simulate revenue and profits based on input of products discount and marketing boost.



5. Loan Pre-Eligibility Scoring

- Random forest model estimates whether a business qualifies for financing.





6. AI Acknowledgement

Tools used:

- **ChatGPT**: Used for drafting report sections, debugging advice (such as pip & numpy issue) and reviewing architecture concepts.

- **GitHub Copilot**: Used inside VS Code to accelerate producing frontend components, API iterators and suggestions for repetitive code blocks.

We declare the use of these AI tools to be transparent. All code, logic and final deliverables were reviewed and adapted by the project team.

7. Conclusion

BizView demonstrates a practical, end-to-end prototype that unites point-of-sale operations, analytics, forecasting, anomaly detection, and an initial loan pre-eligibility engine, it is all tailored for MSMEs. The prototype reduces the time-to-insight for owners and creates a more consistent data package for lenders, helping both sides make better decisions faster. We created a platform that not only helps MSMEs survive daily operations but empowers them to grow, optimize, and secure necessary funding.

Future Roadmap:

Phase 1: Getting Started & Growing Fast (Months 1–6)

- Free Version Launch: We will offer a free version of the basic assessment so anyone can try it out and we can grow our user base quickly.
- BizView Pro Launch: We will release a "Pro" version that turns on advanced AI features and deeper financial tracking for leaders who want more detail.

Phase 2: Expanding Across Southeast Asia (Months 6–18)

- Seamless API links with the most used MSME platforms across all of ASEAN (Grab, TikTok Shop, Shopee, etc.)

Phase 3: Becoming an Industry Leader (Year 2+)

- Connecting MSMEs with partner banks by translating our ML anomaly detection and AI-scoring into high-fidelity risk profiles, enabling frictionless and safe micro-lending.